

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

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8 **ABSTRACT:** When multiple differential cross-section measurements are extracted from a common
9 event sample, the event-sharing structure between distributions imposes exact linear constraints
10 among the bin counts that reduce the effective dimensionality of the measurement below the total bin
11 count. When the statistical covariance matrix accounts for these inter-distribution correlations, the
12 constraints manifest as a rank deficiency. Systematic uncertainties may formally restore invertibility,
13 but the resulting near-singular directions yield unphysically inflated χ^2 values in global goodness-
14 of-fit tests. We present the range-projected χ^2 , a test statistic that restricts the goodness-of-fit
15 test to the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$
16 degrees of freedom, where N_{null} is the number of independent constraints. We derive N_{null} from
17 the bin-combination structure of the measurement, and show how it decomposes into structural
18 contributions determined a priori from the binning geometry and kinematic contributions that
19 depend on the phase-space occupancy. The method is validated with an analytical toy model and a
20 realistic simulated neutrino–argon cross-section measurement including unfolding and multi-source
21 systematic uncertainties.

22 **KEYWORDS:** Analysis and statistical methods

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42 1 Introduction

43 Modern neutrino cross-section literature is experiencing an increase of published measurements
44 reporting multiple differential distributions extracted from the same event sample. MicroBooNE
45 has reported simultaneous measurements of cross sections for final states with and without protons
46 [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables
47 [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple
48 final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from
49 NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple
50 reported distributions, leading to strong inter-measurement statistical correlations. Separate but
51 related efforts have produced simultaneous measurements across different nuclear targets [11–13]
52 and combined neutrino-antineutrino results [14], where correlations arise from shared systematic
53 uncertainties rather than shared events.

54 When multiple distributions are extracted from a common event sample, the event-sharing
55 structure imposes exact linear constraints among the bin counts. If every selected event populates

one bin in each distribution, then the total number of events in all of them is exactly identical. More generally, the pattern of shared bin occupancy across distributions determines a set of constraints that reduce the effective dimensionality of the measurement below the total bin count. These constraints are a property of the measurement itself, independent of any particular statistical treatment.

The blockwise unfolding framework of Ref. [15] provides a rigorous method for computing the full cross-distribution covariance matrices in such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The constraints described above can manifest as a rank deficiency in the statistical covariance matrix, as certain directions in bin space carry identically zero variance. The number of these degenerate directions is exactly predictable from the binning geometry.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, a global χ^2 that does not account for the constrained directions yields a test with the wrong effective number of degrees of freedom. When the statistical covariance matrix is constructed under the Poisson assumption, the constraints appear as exact null directions that prevent direct matrix inversion. Systematic uncertainties may formally restore invertibility, but the resulting near-singular directions produce unphysically inflated χ^2 values. In either case, the test is unreliable unless the constrained directions are properly identified and excluded.

We present a method to correctly compute a global goodness-of-fit test statistic for multi-distribution measurements with shared events. The approach is based on identifying the constrained directions from the bin structure of the measurement and projecting the statistic onto the subspace where independent statistical information is available. We refer to the resulting test statistic as the range-projected χ^2 . We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

2 Constraints from shared events

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common event sample then each selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical

97 covariance matrices that respect the correlations between bins in different distributions or blocks.
 98 The full inter-block covariance matrix constructed in reconstructed space is propagated through
 99 unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual un-
 100 folding matrices computed for each block. The result of the combined measurement is a vector of
 101 $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix. The total covariance
 102 decomposes as $C_{\text{total}} = C_{\text{stat}} + C_{\text{sys}}$, where the statistical covariance component, computed from the
 103 shared-event structure, encodes the constraints described in the introduction.

104 2.1 Statistical covariance from shared events

105 The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach
 106 is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-
 107 overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin
 108 u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to
 109 the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each
 110 bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the
 111 counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (2.1)$$

112 as by definition bins u , v , and w have no events in common. This implies that the covariance
 113 between any two bins can be estimated from the variance of the events they share.

114 The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their
 115 squared weights. This gives us a general prescription to calculate the elements of the statistical
 116 covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2.2)$$

117 where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the
 118 case of measured data the weights are 1, and the covariance element simplifies to the number of
 119 events shared between the bins n_{ij} .

120 After propagation through the block-diagonal unfolding, the statistical covariance matrix in
 121 unfolded space inherits the same rank structure, as the unfolding matrices for each individual block
 122 are invertible. This rank structure has a direct interpretation in terms of the number of distinct
 123 event-sharing patterns across distributions, which we discuss next.

124 2.2 Combination matrix and null space

125 The previous discussion shows that the statistical covariance between any two bins is determined
 126 solely by the events they share. We now formalize this observation to derive an exact formula for
 127 the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical
 128 variance.

129 For each event e , we can define the indicator vector $\phi_e := (\phi_{e,i})_{i=1}^{N_{\text{bins}}}$ such that $\phi_{e,i} = 1$ if
 130 $i \in \text{bins}(e)$ and $\phi_{e,i} = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all

131 distributions. The indicator vector acts as an identifier of each event in the combined measurement.
 132 For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1
 133 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix
 134 can then be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^T. \quad (2.3)$$

135 This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the
 136 statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct
 137 indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is
 138 removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

139 We define the bin-combination matrix Ω as the matrix whose rows are the unique indicator
 140 vectors among $\{\phi_e\}$. Each row of Ω represent a distinct pattern of bin occupancy across the blocks.
 141 By construction the row space of Ω equals $\text{span}\{\phi_e\}$, and therefore:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(\Omega). \quad (2.4)$$

142 The number of degenerate directions is then:

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega). \quad (2.5)$$

143 Physically, each of the N_{null} directions corresponds to a linear combination of bins along which
 144 the measurement has identically zero variance. Any component of the residual vector along such
 145 a direction is not perceived by the statistical covariance; it is a direction that the measurement has
 146 no statistical power to constrain. Importantly, both Ω and N_{null} are directly computable from the
 147 MC event-to-bin mapping alone, before constructing any covariance matrix, confirming that the
 148 constraints are a property of the measurement geometry rather than the statistical treatment.

149 A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

150 or equivalently $\Omega \mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space
 151 vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different
 152 events. The null space directions can be divided into two physically distinct categories: structural
 153 constraints that follow from the block topology alone, and kinematic constraints that depend on
 154 which regions of phase space are populated.

155 **2.3 Determination of the structural null space**

156 The combination matrix Ω presented previously is constructed from MC events. One determines
 157 which events populate the different bins and collects the unique indicator vectors. However, the
 158 indicator vectors that can appear are constrained by two distinct mechanisms. The first is purely
 159 structural, the geometry of the binning scheme forbids certain bin combinations. The second is
 160 kinematic, correlations between the different variables render some combinations unreachable even
 161 though they are geometrically allowed. These constraints are imposed by the underlying physics

162 effects, mainly the differential cross section and the detector acceptance. In the following, we show
 163 that the structural constraints can be determined a priori, without generating MC events, from the
 164 bin edges used.

165 The variables used in a measurement are drawn from a set $\{v_1, \dots, v_D\}$. Each block b uses
 166 a subset of these variables, $\text{vars}(b) \subseteq \{v_1, \dots, v_D\}$. Restricting the discussion to combinations of
 167 single- and double-differential measurements, we have $|\text{vars}(b)| \in \{1, 2\}$ for every b . For each
 168 variable v used in a given block b , the binning scheme defines a set of bin edges $E_v^{(b)}$ that partition
 169 the range of v into intervals. In the case of a two-dimensional block, the edges in the binning
 170 variable may be slice-dependent, but nevertheless these can be collected into a single set of edges
 171 per variable across slices. For each variable $v \in \bigcup_b \text{vars}(b)$, we can define the refined edges:

$$\mathcal{E}_v = \bigcup_{b: v \in \text{vars}(b)} E_v^{(b)}, \quad (2.7)$$

172 with $M_v = |\mathcal{E}_v| - 1$ denoting the number of refined bins for variable v . Since $\mathcal{E}_v$ contains all the
 173 bin edges from every block that uses that variable, each refined bin is contained entirely within one
 174 original bin of every such block.

175 The common refinement of all blocks is the Cartesian product of the refined bins across all
 176 variables, denoted by $\mathcal{R}$. Each cell $c \in \mathcal{R}$ specifies a refined bin index for every variable in the
 177 measurement. For every block b , there exists a unique mapping $\beta_b(c)$ between each cell c and an
 178 original bin in said block. The elements of the indicator vector $\phi_c := (\phi_{c,i})_{i=1}^{N_{\text{bins}}}$ associated to cell
 179 c can be written as:

$$\phi_{c,i} = \begin{cases} 1 & \text{if } i = \beta_b(c) \text{ for any block } b, \\ 0 & \text{otherwise.} \end{cases} \quad (2.8)$$

180 If block b does not use variable v then β_b is independent of the refined bin index in v . Therefore,
 181 multiple cells may produce identical indicator vectors.

182 The structural combination matrix Ω_{struct} is defined as the matrix whose rows are the unique
 183 indicator vectors from the cells of the common refinement $\{\phi_c : c \in \mathcal{R}\}$. By construction, every
 184 indicator vector that appears in the Ω matrix obtained from the MC events is also present in Ω_{struct} ,
 185 therefore:

$$\text{rank}(\Omega) \leq \text{rank}(\Omega_{\text{struct}}) \leq N_{\text{bins}}. \quad (2.9)$$

186 This hierarchy motivates a decomposition of the total number of degenerate directions:

$$\begin{aligned} N_{\text{null}} &= N_{\text{bins}} - \text{rank}(\Omega_{\text{struct}}) + \text{rank}(\Omega_{\text{struct}}) - \text{rank}(\Omega) \\ &= N_{\text{null}}^{\text{struct}} + N_{\text{null}}^{\text{kinem}}. \end{aligned} \quad (2.10)$$

187 The first term, $N_{\text{null}}^{\text{struct}}$, counts null directions that arise purely from the binning geometry and is
 188 computable a priori from the bin edges alone. The second term, $N_{\text{null}}^{\text{kinem}}$, accounts for additional null
 189 directions introduced by phase-space restrictions that exclude geometrically allowed combinations
 190 and can only be determined from Monte Carlo. A description of the algorithm used to compute
 191 Ω_{struct} is presented in Appendix A.

192 The presence of kinematic null directions indicate that certain regions of the combined bin
 193 space are empty, which may signal that the binning extends beyond the kinematic reach of the

measurement or that it is finer than the available statistics can support. In such cases, merging or removing the offending bins may be preferable to relying on a projection to handle the resulting constraints. The structural null directions, by contrast are an unavoidable consequence of measuring multiple distributions from shared events and cannot be eliminated by rebinning.

The structure of the null space of Ω_{struct} depends on how the bin edges for the different blocks overlap. In the simplest case, when blocks do not share variables, every combination of one bin per block is allowed. The resulting null space is $(B - 1)$ -dimensional, with a single constraint that requires all block totals to be equal. When blocks share one or more variables, the allowed combinations are restricted and the dimension of the null space can be different from $B - 1$. The number of null directions depends on how shared variable bin edges are aligned between blocks. Aligned slice edges decouple the measurement into different sub-problems, adding additional sum rules and increasing $N_{\text{null}}^{\text{struct}}$. On the other hand, misaligned edges couple adjacent distributions, potentially reducing the number of constraints. In the general case, the interplay between shared variables, bin edge alignment, and block topology makes a closed-form expression impractical, but the computation of $\text{rank}(\Omega_{\text{struct}})$ from the bin edge specifications is straightforward.

The structural and kinematic components of the null space respond differently to systematic uncertainties. Whether these cause the rank deficiency of C_{stat} to persist, partially resolve, or fully resolve in the total covariance C_{total} depends on the nature of the systematic treatment, which we discuss next.

2.4 Effect of systematic uncertainties

Previously, we established that the event-sharing structure imposes $N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega)$ exact linear constraints on the bin counts, which manifest as a rank deficiency of C_{stat} . Whether the total covariance matrix inherits the rank deficiency depends on which of these null directions receive support from the systematic uncertainties. Now, we examine the mechanisms by which different systematic treatments do or do not populate the null space of C_{stat} .

A systematic variation s produces a modified vector of expected events $\mathbf{n}_s$, whose departure from the central-value prediction can be expressed in terms of a shift vector:

$$\delta_s = \mathbf{n}_s - \mathbf{n}_{\text{CV}}. \quad (2.11)$$

The different contributions to C_{syst} are constructed from the outer product of these shift vectors. A null direction $\mathbf{x}$ of C_{stat} remains a null direction of C_{total} if and only if $\mathbf{x} \cdot \delta_s = 0$ for every systematic variation in the uncertainty budget. The question is therefore whether the shift vectors have nonzero projections onto the structural and kinematic null spaces.

Consider first the simplest case, where a systematic s acts by reweighting the reconstructed event counts directly. Each event e is assigned a modified weight $w_e^{(s)}$ while preserving its bin assignments. The shift vector can then be written as:

$$\delta_s = \sum_e (w_e^{(s)} - w_e) \phi_e. \quad (2.12)$$

For any null vector $\mathbf{x}$ which satisfies the null-space condition of Eq. (2.6), the projection of $\mathbf{x}$ onto the shift vector gives:

$$\begin{aligned}\mathbf{x} \cdot \boldsymbol{\delta}_s &= \sum_e (w_e^{(s)} - w_e) (\boldsymbol{\phi}_e \cdot \mathbf{x}) \\ &= \sum_e (w_e^{(s)} - w_e) \sum_{i=1}^{N_{\text{bins}}} \phi_{e,i} x_i \\ &= 0,\end{aligned}\tag{2.13}$$

since the inner sum vanishes for every event. Direct reweighting therefore preserves the entire null space, adding no rank to C_{total} . Flux uncertainties, which modify the neutrino energy spectrum through per-event weights, are the classic example. Cross-section model variations and hadronic reinteraction uncertainties, when implemented as per-event reweighting, behave identically.

A qualitatively different situation arises when the systematic uncertainties are propagated through the response matrix rather than changing the event counts per bin directly. In this approach, adopted as the standard prescription in MicroBooNE for neutrino interaction uncertainties [15], a modified response matrix Δ'_s , mapping true bins to reconstructed bins under the given variation, is constructed for each systematic uncertainty. This matrix is obtained either by reweighting the events used to build the nominal response matrix or from an independent MC sample with varied simulation parameters. In this case, the predicted reconstructed counts for systematic variation s are:

$$\mathbf{n}_s = \Delta'_s \boldsymbol{\varphi}_{\text{CV}},\tag{2.14}$$

where $\boldsymbol{\varphi}_{\text{CV}}$ is the central-value truth prediction. The shift vector is then:

$$\boldsymbol{\delta}_s = (\Delta'_s - \Delta) \boldsymbol{\varphi}_{\text{CV}}.\tag{2.15}$$

Since Δ is block-diagonal, the difference $\Delta'_s - \Delta$ acts independently on each block, projecting the nominal truth vector through block-specific changes in efficiency and migration. The per-event coupling that makes direct reweighting respect all null directions is absent in this prescription. When the weight of an event changes, it affects simultaneously every block it populates; the response-matrix method, in contrast, processes each block independently through its own sub-matrix. The projection of $\boldsymbol{\delta}_s$ onto any null vector, whether structural or kinematic, therefore involves differences in how each block responds to the same truth-level prediction, and in general is nonzero. Response-matrix variations thus break both types of null directions, regardless of how Δ'_s is obtained.

A third mechanism applies when systematic uncertainties are evaluated by directly comparing reconstructed event counts from independent MC samples simulated with nominal and varied parameters. In this case, the shift vector takes the form:

$$\boldsymbol{\delta}_s = \sum_{e'} w_{e'}^{(s)} \boldsymbol{\phi}_{e'}^{(s)} - \sum_e w_e \boldsymbol{\phi}_e,\tag{2.16}$$

where the first sum runs over the events in the varied-model sample and the second sum over the nominal sample. Since each event in both samples populates exactly one bin per block, their difference projects to zero onto the structural null directions by an analogous argument to that used for the direct reweighting. Kinematic null directions, however, can be lifted if the

variation scatters events into reconstructed-bin combinations absent in the nominal sample. In that case, new indicator vectors appear that are not rows of Ω , potentially breaking the corresponding constraints. Detector systematic uncertainties, which usually require independent samples with modified simulation parameters that cannot be captured by event reweighting, are the typical example where this prescription is applied.

In practice, whether the eigenvalues of C_{total} along the formerly null directions are exactly zero or merely smaller than the physical eigenvalues, including these directions in a χ^2 test via standard matrix inversion produces inflated values with the wrong effective number of degrees of freedom. A valid goodness-of-fit test must therefore be restricted to the subspace where the measurement carries genuine sensitivity, which is the role of the projection method developed in the following section.

3 Range-projected χ^2 test statistic

The discussion presented in Sec. 2 established that the event-sharing structure of the multi-distribution measurement imposes N_{null} linear constraints, encoded in the null space of the combination matrix Ω , which emerges as a rank deficiency in C_{stat} . A global goodness-of-fit test statistic must therefore be formulated in the subspace orthogonal to these constrained directions, where independent statistical information is available.

The constraints are most naturally identified in reconstructed space, where the combination matrix is constructed. The non-null subspace can be obtained decomposing $\Omega^T \Omega$. This matrix is a square $N_{\text{bins}} \times N_{\text{bins}}$ matrix with the same column space and null space as Ω itself, making it a more compact representation of the same information. We can write its eigendecomposition as:

$$\Omega^T \Omega = V S V^T, \quad (3.1)$$

where $S = \text{diag}(s_1, \dots, s_{N_{\text{bins}}})$ contains the eigenvalues in decreasing order and V is the orthogonal matrix of eigenvectors. Of these eigenvalues, $r = \text{rank}(\Omega)$ are nonzero. Partitioning $V = [V_r \mid \tilde{V}_r]$ into the columns associated with the nonzero and zero eigenvalues, the columns of V_r span the subspace of reconstructed bin space that is not constrained by the event-sharing structure. The orthogonal projector onto this subspace is:

$$P_{\text{reco}} = V_r V_r^T. \quad (3.2)$$

Note that P_{reco} is identical to the projector onto the range of C_{stat} as defined in Eq. (2.3), since C_{stat} and $\Omega^T \Omega$ share the same column space by construction. The combination matrix formulation makes explicit the fact that the projector is determined by the measurement geometry.

Cross-section measurements are reported in unfolded space, and it is in that space where goodness-of-fit tests are typically evaluated. The blockwise unfolding matrix U , which is block-diagonal and invertible, maps reconstructed-space vectors onto unfolded space. The non-null basis vectors propagate as $W_r = U V_r$. Since U is not orthogonal in general, the columns of W_r are not orthonormal, and the projector onto their span takes the general form:

$$P_{\text{true}} = W_r (W_r^T W_r)^{-1} W_r^T. \quad (3.3)$$

This projector identifies the subspace of unfolded bin space that carries independent statistical information, with detector response and unfolding effects properly accounted for. Since U is invertible, W_r has the same rank as V_r and the number of constrained directions is preserved. A test statistic constructed from the projected covariance matrix would have $r = N_{\text{bins}} - N_{\text{null}}$ effective degrees of freedom.

In practice, the eigenvalues of $\Omega^T \Omega$ that are technically zero take small numerical values. The threshold for identifying these can be set relative to the largest eigenvalue (e.g., $s_i/s_1 < 10^{-10}$). In most cases, a clear spectral gap of several orders of magnitude separates the physical eigenvalues from the numerical noise, making the identification unambiguous.

3.1 Test statistic definition

The goodness-of-fit test is evaluated in reconstructed space, where the residual vector is given by $\delta = \mathbf{d} - \Delta \boldsymbol{\varphi}_{\text{pred}}$, where $\mathbf{d}$ is the background-subtracted data, Δ is the detector response matrix, and $\boldsymbol{\varphi}_{\text{pred}}$ is a truth-level model prediction. The range-projected χ^2 is defined as:

$$\chi_{\text{proj}}^2 = \delta^T (P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ \delta, \quad (3.4)$$

with the superscript $+$ denoting the Moore–Penrose pseudoinverse.

The pseudoinverse can be avoided entirely by reducing the problem to r dimensions. One can verify that the pseudoinverse of $P_{\text{reco}} C_{\text{total}} P_{\text{reco}}$ takes the closed form:

$$(P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ = V_r (V_r^T C_{\text{total}} V_r)^{-1} V_r^T. \quad (3.5)$$

Substituting into Eq. (3.4) and defining the reduced residual $\tilde{\delta} = V_r^T \delta$ and the reduced covariance matrix $\tilde{C}_{\text{total}} = V_r^T C_{\text{total}} V_r$, the test statistic becomes:

$$\chi_{\text{proj}}^2 = \tilde{\delta}^T \tilde{C}_{\text{total}}^{-1} \tilde{\delta}, \quad (3.6)$$

which involves only a standard inversion of the $r \times r$ matrix $\tilde{C}_{\text{total}}$. Under the null hypothesis, χ_{proj}^2 follows a χ^2 distribution with $r = N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

Without the projection, the χ^2 test statistic takes the same numerical value whether computed in reconstructed or unfolded space, as the factors of the unfolding matrix cancel identically when applying the additional smearing matrix to the predictions [17]. The introduction of the projector breaks this equivalence when the projection is performed after unfolding. In unfolded-space the residual is defined as $\mathbf{r} = \hat{\boldsymbol{\varphi}} - A_C \boldsymbol{\varphi}_{\text{pred}}$, where $\hat{\boldsymbol{\varphi}} = U \mathbf{d}$ is the unfolded background-subtracted data and $A_C = U \Delta$ is the additional smearing matrix. Using the propagated basis $W_r = U V_r$, one can construct an analogous test statistic in unfolded space as:

$$\chi_{\text{unf}}^2 = \tilde{\mathbf{r}}^T \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}}, \quad (3.7)$$

where $V_{\text{total}} = U C_{\text{total}} U^T$ is the total covariance matrix in unfolded space, and $\tilde{\mathbf{r}} = W_r^T \mathbf{r}$ and $\tilde{V}_{\text{total}} = W_r^T V_{\text{total}} W_r$ are the . While this is a valid test statistic on its own, it is not numerically equal to Eq. (3.6). Expressing it in terms of reconstructed-space quantities yields:

$$\chi_{\text{unf}}^2 = \delta^T U^T U V_r (V_r^T U^T U C_{\text{total}} U^T U V_r)^{-1} V_r^T U^T U \delta, \quad (3.8)$$

which reduces to Eq. (3.6) only when $U^T U = I$, i.e. when the unfolding matrix is orthogonal.

The χ^2 equivalence is restored if the projection is applied before unfolding. Projecting the reconstructed-space residual and covariance first, then propagating through the unfolding matrix, gives:

$$\begin{aligned} (U P_{\text{reco}} \delta)^T (U P_{\text{reco}} C_{\text{total}} P_{\text{reco}} U^T)^+ (U P_{\text{reco}} \delta) &= \tilde{\delta}^T W_r^T (W_r \tilde{C}_{\text{total}} W_r^T)^+ W_r \tilde{\delta} \\ &= \tilde{\delta}^T G (G \tilde{C}_{\text{total}} G)^{-1} G \tilde{\delta} \\ &= \chi_{\text{proj}}^2, \end{aligned} \quad (3.9)$$

where the symmetric matrix $G = W_r^T W_r$ acts as the unfolding matrix of the reduced system, and cancels similarly to the unprojected case. In the approach of Eq. (3.8), the full residual, including the null-space components, is first rotated by U into directions that are no longer orthogonal to the range. The unfolded-space projector then picks up this null-space contamination. Projecting first in reconstructed space avoids this by removing the null directions before unfolding can mix them with the physical subspace.

For this reason, Eq. (3.6) is the recommended definition of the range-projected χ^2 .

3.2 Statistical validity

Under the null hypothesis, the residual $\mathbf{r}$ is drawn from a multivariate Gaussian distribution with mean zero and covariance V_{total} . The reduced residual $\tilde{\mathbf{r}} = W_r^T \mathbf{r}$ is a linear transformation of a Gaussian vector, therefore Gaussian itself with mean zero and covariance $\tilde{V}_{\text{total}} = W_r^T V_{\text{total}} W_r$. Provided that $\tilde{V}_{\text{total}}$ is positive definite, guarantees that $\tilde{\mathbf{r}}^T \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}} \sim \chi^2(r)$. The positive definiteness of $\tilde{V}_{\text{total}}$ requires V_{total} to have nonzero variance along every direction in the range of the projector P_{true} , which is generically satisfied. This can be verified by checking that all eigenvalues of $\tilde{V}_{\text{total}}$ are strictly positive.

The validity of the range-projected χ^2 can also be understood from the general theory of quadratic forms of correlated normal variables. Theorem 9.2.2 of Ref. [22] establishes that if $\mathbf{Y} \sim \mathcal{N}(0, \Sigma)$ with $\text{rank}(\Sigma) = k$, then $\mathbf{Y}^T \Sigma^+ \mathbf{Y} \sim \chi^2(k)$ for any choice of generalized inverse Σ^+ . Assuming the null hypothesis, the projected residual $P_{\text{true}} \mathbf{r}$ is normally distributed with covariance $P_{\text{true}} V_{\text{total}} P_{\text{true}}$, which has rank r by construction. The formulation of Eq. (3.4) with the Moore–Penrose pseudoinverse is a specific instance of this general result. The key distinction from purely numerical approaches to handling singular covariance matrices is that the rank is not determined by an eigenvalue threshold applied to the covariance, but is predicted exactly via the combination matrix Ω from the structure of the measurement.

3.3 Relation to alternative approaches

4 Toy example

The formalism developed previously is now illustrated with a toy model that captures the essential features of a multi-distribution neutrino cross-section measurement: a two-dimensional underlying physics process, a simple detector model, and a multi-block binning scheme in which events are shared across distributions. The toy is design to be minimal but non-trivial. A three-block structure produces a rank-deficient statistical covariance matrix, and a Feldman–Cousins study demonstrates the consequences of ignoring or correctly treating the resulting null directions.

4.1 Model and measurement setup

The toy model mimics a muon neutrino charged-current quasi-elastic-like interaction, producing a muon characterized by its momentum p_μ and scattering angle θ_μ . The joint truth-level differential cross section is parametrized as:

$$f(p_\mu, \cos \theta_\mu) = p_\mu^k e^{-k p_\mu/p_0} e^{\kappa \cos \theta_\mu} (1 + \alpha p_\mu \cos \theta_\mu), \quad (4.1)$$

where the four parameters (k, p_0, κ, α) control the shape of the momentum spectrum, the level of forward peaking, and the strength of the momentum-angle correlation, respectively. A central-value (CV) model with $(k, p_0, \kappa, \alpha) = (3.0, 0.2, 2.0, 0.3)$ is used to build the response matrix and estimate the covariance, while a “fake” model with $(k, p_0, \kappa, \alpha) = (3.3, 0.2, 1.8, 0.35)$ serves as the true underlying distribution from which pseudo-data are drawn. Truth events are generated via rejection sampling over $p_\mu \in [0.0, 1.2]$ GeV/ c and $\cos \theta_\mu \in [-1.0, 1.0]$.

A simple detector model introduces two different effects. Momentum is smeared multiplicatively, $p_{\text{reco}} = p_{\text{true}}(1 + \delta_p)$ with $\delta_p \sim \mathcal{N}(0, \sigma_p)$ and $\sigma_p = 0.05$, while the angle is smeared additively, $\theta_{\text{reco}} = \theta_{\text{true}} + \delta_\theta$ with $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta)$ and $\sigma_\theta = 1^\circ$. A sigmoid efficiency function $\varepsilon(p_{\text{reco}}) = [1 + e^{-(p_{\text{reco}} - p_{\text{thr}})/w}]^{-1}$ with threshold $p_{\text{thr}} = 0.10$ GeV/ c and width $w = 0.02$ GeV/ c models the turn-on of the event acceptance at low momentum.

The analysis uses three bin blocks in reconstructed space: a p_μ marginal (5 bins), a $\cos \theta_\mu$ marginal (5 bins), and a two-dimensional distribution sliced in $\cos \theta_\mu$ (3 slices, each with 4 bins in p_μ), for a total of $N_{\text{bins}} = 22$. The bin edges of the 2D block do not align exactly with those of the 1D marginals, which is the usual situation in real analyses. A CV sample of $N_{\text{MC}} = 500,000$ truth events is generated, processed through the detector model, and used to construct the response matrix Δ . Pseudo-data samples are generated by drawing $N_{\text{data}} = 5,000$ events from the fake model.

Systematic uncertainties are estimated following the response-matrix approach described in Sec. 2.4. Interaction-model variations are obtained by reweighting the CV sample to six alternative model configurations, each producing a varied response matrix Δ_s . The shift vector $\delta_s = (\Delta_s - \Delta) \varphi$ captures the effect on the predicted reconstructed-space distribution. Detector variations are obtained by generating eight independent samples with perturbed detector parameters (σ_p , σ_θ , p_{thr} , and w), each producing a new response matrix from which a shift vector is computed similarly. The total systematic covariance matrix is $C_{\text{syst}} = N_{\text{var}}^{-1} \sum_i \delta_i \delta_i^T$, scaled to data statistics. The statistical covariance is computed using Eq. (2.2), assuming unit event weights, which captures the inter-block correlations from event sharing.

Figure 1 shows the predicted event distributions for the CV and fake models in all three blocks of the toy analysis, together with a pseudo-data sample distribution and the systematic uncertainty bands.

4.2 Null-space structure

The structural combination matrix Ω_{struct} is constructed from the binning configuration of the three blocks, following the prescription from Sec. 2.3. For this bin scheme, Ω_{struct} is rank 17, yielding 5 structural null directions. The combination matrix Ω computed from the MC event list also has rank 17, confirming that all 5 null directions are structural. No additional kinematic constraints arise for this choice of physics model and binning.

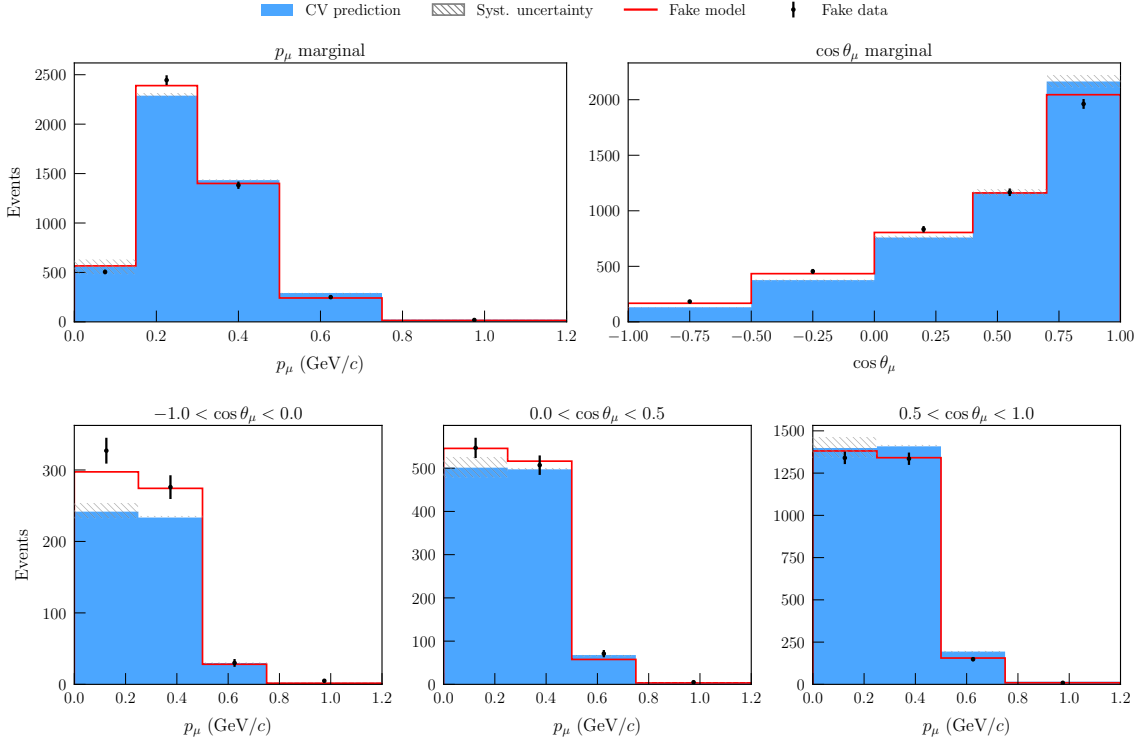


Figure 1. Predicted event distributions in the toy analysis blocks for the CV model (filled histograms) and the fake model (solid lines), together with a pseudo-data realization (points with statistical error bars). The shaded bands indicate the systematic uncertainty on the CV prediction. The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block, each binned in p_μ .

The eigenvalue spectrum of C_{stat} confirms this. It shows 5 eigenvalues that are numerically zero, separated from the 17 nonzero eigenvalues by a spectral gap of several orders of magnitude. Adding systematic uncertainties constructed via the response-matrix approach lifts the 5 null eigenvalues, making $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ formally full rank. However, as shown in Fig. 2, the lifted eigenvalues remain considerably smaller than the bulk of the spectrum. Along these directions the measurement carries no independent statistical information, therefore the variance is determined entirely by the systematic model. Including them in the test statistic introduces poorly calibrated contributions to the χ^2 , as demonstrated next.

4.3 Goodness-of-fit validation

The calibration of the projected and unprojected χ^2 test statistics is assessed with a toy Monte Carlo exercise at the true hypothesis point. A high-statistics simulation of the fake model is used to estimate the event distribution on a fine reconstructed-space grid, defined by the union of all bin edges in p_μ and $\cos \theta_\mu$ across the three blocks, corresponding to the common refinement described in Sec. 2.3. Each toy experiment draws a Poisson-fluctuated total event count and distributes them across the fine grid via multinomial sampling. These events are then projected onto the three analysis blocks, preserving the inter-block correlations from event sharing. Systematic fluctuations are added as correlated Gaussian throws from the Cholesky decomposition of C_{syst} . The χ^2 is evaluated in

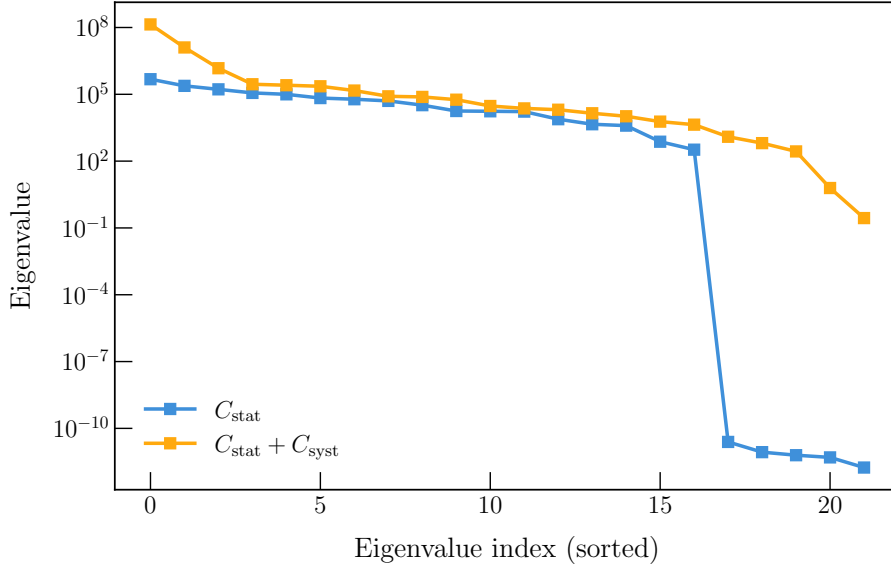


Figure 2. Eigenvalue spectra of the statistical covariance matrix C_{stat} (blue circles) and the total covariance matrix $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ (orange squares), for the toy analysis, sorted in descending order.

reconstructed space, comparing each toy to the forward-folded prediction at the true hypothesis, for both the unprojected statistic, using the direct inverse of C_{total} , and the range-projected statistic of Eq. (??).

Figure 3 shows the results from 10^6 toy experiments. The projected χ^2 statistic closely follows the expected $\chi^2(17)$ distribution, with a mean of 17.23 and a standard deviation of 6.01, in good agreement with the expected values of 17 and $\sqrt{2 \times 17} \approx 5.83$. In contrast, the unprojected statistic is shifted to significantly higher values with a mean of 30.59, and a standard deviation of 8.96, far above the expectation for a $\chi^2(22)$ distribution of 22 and $\sqrt{2 \times 22} \approx 6.63$. The right panel shows the corresponding p-value distributions. The projected statistic yields an approximately uniform distribution, with a false rejection rate of 5.7% at the 5% significance level. The unprojected statistic produces a false rejection rate of 32.4%, more than six times the expected value. An analyzer using the unprojected statistic for goodness-of-fit testing would falsely reject the correct model roughly one third of the time.

4.4 Parameter estimation

The impact of using the projected test statistic on parameter estimation is assessed with a two-dimensional Feldman–Cousins study. A grid of hypotheses is constructed over (κ, α) with $k = 3.3$ held constant. At each grid point, the truth-level prediction is forward-folded through the nominal response matrix and compared to the toy pseudo-data. The test statistic $\Delta\chi^2 = \chi^2(\kappa, \alpha) - \chi_{\text{min}}^2$ is computed for both the unprojected and projected cases, and confidence regions are constructed from the resulting $\Delta\chi^2$ surfaces.

Figure 4 shows the 68% and 95% confidence level contours from the Asimov dataset for both test statistics. The projected contours are correctly centered on the true parameter values, while the unprojected contours are shifted, with the 68% region excluding the true point. The unprojected

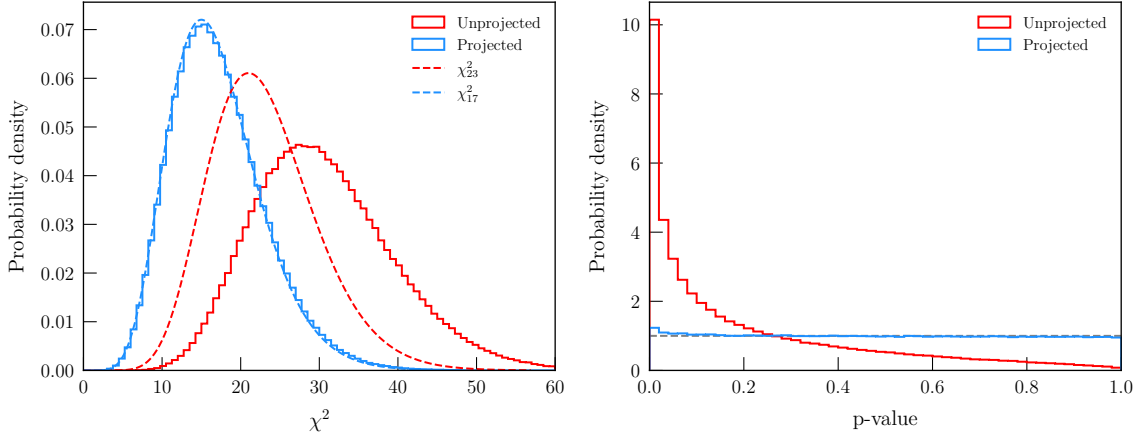


Figure 3. Goodness-of-fit validation for toy experiments generated at the true hypothesis. Left: distributions of the unprojected (red) and range-projected (blue) χ^2 statistics, compared with the expected $\chi^2(22)$ and $\chi^2(17)$ distributions (dashed curves). Right: corresponding p-value distributions, with the false rejection rates at the 5% significance level indicated.

best fit values are biased, particularly in α . The mean best-fit value across toys is $\hat{\alpha} = 0.22$, far from the true value of 0.35, while the projected statistic recovers $\hat{\alpha} = 0.37$. The bias in κ is milder ($\hat{\kappa} = 1.84$ for unprojected, $\hat{\kappa} = 1.80$ for projected, compared to the true 1.80). This indicates that α has greater overlap with the null directions.

The frequentist coverage confirms these observations. At the 68.3% confidence level, the projected statistic achieves nominal coverage in both the joint two-dimensional region (68.2%) and the one-dimensional profiled intervals (68.1% for κ , 67.9% for α). The unprojected statistic suffers severe undercoverage: 36.5% in the joint region, 44.6% for κ , and 34.1% for α . The undercoverage is worst for α , consistent with its stronger sensitivity to the null-space contamination. The projected contours are larger than the unprojected ones, reflecting the genuine sensitivity of the measurement once the ill-constrained directions are properly removed.

5 Realistic example

Acknowledgments

A Structural combination matrix algorithm

We describe the algorithm implemented to construct the structural combination matrix Ω_{struct} from the bin edge specifications of the multi-block measurement. Each row of Ω_{struct} is a unique indicator vector encoding one structurally allowed combination of bins, one per block, that a single event can simultaneously populate. The rank of this matrix determines the number of null directions, as described in Sec. 2.3.

The construction proceeds in three stages. First, the kinematic variables used across all blocks are classified as shared (appearing in two or more blocks) or private (appearing in exactly one block). This distinction is key to the efficiency of the algorithm. Shared variables create

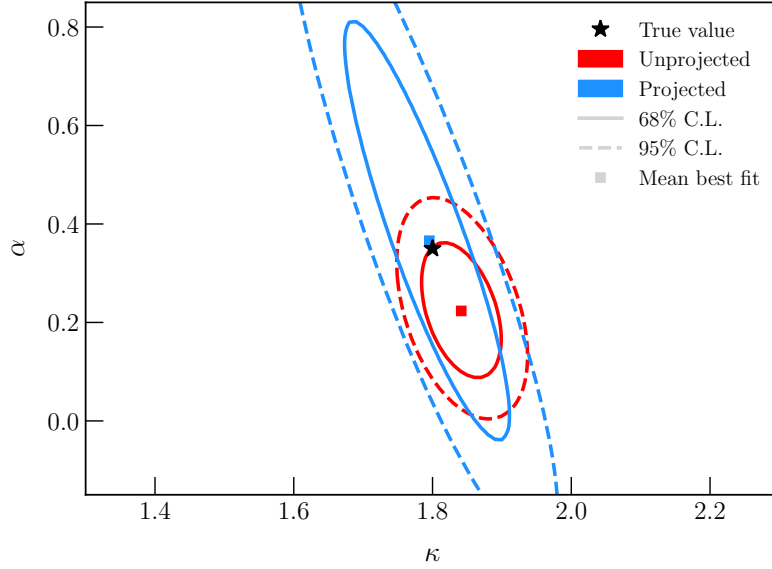


Figure 4. Confidence regions in the (κ, α) plane from the Asimov dataset, showing the 68% and 95% CL contours for the range-projected (blue) and unprojected (red) test statistics. The star marks the true parameter values; the squares indicate the mean best-fit points across 10^6 toys.

459 correlations between blocks, while private variables contribute bins that are independent of all
 460 other blocks.

461 Second, a common refinement is computed for each shared variable by taking the union of all
 462 bin edges from every block that uses it. The algorithm then enumerates the cells of this shared-
 463 variable refinement. For each cell, the midpoint in each shared variable is used to identify which
 464 bin (or set of bins) in each block is compatible with that cell. When the binning of a given block is
 465 fully determined by the shared variables (e.g., a 1D block whose variable is shared, or a 2D block
 466 whose slice and bin variables are both shared) the cell maps to a single bin. When a block uses
 467 a private variable, the shared cell constrains only part of the binning, and the block contributes
 468 multiple candidate bins corresponding to the different private-variable choices. A 1D block whose
 469 variable is private contributes all of its bins; a 2D block with a shared slice variable and a private
 470 bin variable contributes all bins within the determined slice.

471 Third, the candidate bins from each block are combined. In the simplest case, the full Cartesian
 472 product of all candidate bins is expanded, and each element becomes a row of Ω_{struct} . However,
 473 when many blocks have private variables, this product can become very large. In that case,
 474 a representative subset of indicator vectors is used instead. A base combination is formed by
 475 selecting the first candidate bin from each block, and then the remaining candidates from each block
 476 are substituted one at a time while the others are held at their base values. This produces at most
 477 $1 + \sum_b (n_b - 1)$ rows per shared cell, where n_b is the number of candidate bins for block b . The
 478 representative set spans the same column space as the full Cartesian product, and therefore yields
 479 the same $\text{rank}(\Omega_{\text{struct}})$ and null-space structure.

480 To understand why, note that any indicator vector from the full product can be written as the
 481 sum of indicator vectors from the representative set. Denoting the base combination as $\mathbf{e}_0$ and the

single-block variations as $\mathbf{e}_j^{(k)}$ (block k switched to its j -th candidate), any full-product row that selects candidate j_k in each block satisfies:

$$\mathbf{e}_{j_1, \dots, j_B} = \mathbf{e}_0 + \sum_k (\mathbf{e}_{j_k}^{(k)} - \mathbf{e}_0), \quad (\text{A.1})$$

since the blocks occupy disjoint columns. The full product therefore lies in the span of the representative rows, and the rank is preserved.

When the measurement involves mutually exclusive event selections, blocks belonging to different selections share no events and are processed independently. The algorithm is applied separately to each selection group, and the resulting matrices are vertically concatenated.

The cost of the algorithm is dominated by the enumeration of shared-variable cells, which scales as $\prod_{v \in \text{shared}} M_v$ where M_v is the number of refined bins for shared variable v . Since only shared variables enter this product, measurements with many blocks but few shared variables, such as a set of 1D distributions in different kinematic variables, are handled efficiently regardless of the total number of bins.

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